Hyungjin Chung, Ph.D.

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Research interests

Generative models, Alignment, Imaging, AI for medicine

Work Experience

Korea University *Seoul, Korea*
Assistant Professor, Dept. of CSE 2026.09 — Current

EverEx *Seoul, Korea*
Director of Resesarch 2026.09 — Current
Lead AI Research Scientist 2024.08 — 2026.08

NVIDIA Research *San Jose, USA (remote)*
Research Scientist Intern, AI4Science 2023.11 — 2024.01

Google Research *Mountain View, USA*
Student Researcher, team LUMA 2023.07 — 2023.10

Los Alamos National Laboratory *Los Alamos, USA*
Research intern, Applied math & Plasma physics 2022.06 — 2022.08

Education

KAIST *Daejeon, Korea*
Ph.D., Bio & Brain Engineering 2019.03 — 2025.02
Advisor: Jong Chul Ye
Thesis: *Practical approximations of posterior sampling in diffusion model-based inverse problems*

Korea University *Seoul, Korea*
B.S., Biomedical Engineering 2015.03 — 2019.02

Conf. publications

*First authors, †Corresponding authors. If absent, first/last authors are first/corresponding.

[C31] Align Your Query: Representation Alignment for Multimodality Medical Object Detection
Ara Seo*, Bryan Sangwoo Kim*, Hyungjin Chung, Jong Chul Ye
WACV 2027

[C30] MVVBench: Benchmarking 4D Reasoning in Vision-Language Models
Hyungjin Chung*, Byeongjun Park*, Joonseok Lee, Hojun Kim, Jaeho Choi, Byung-Hoon Kim
NeurIPS 2026 Evaluations & Datasets

[C29] Accelerating Video Inverse Problem Solvers with Autoregressive Diffusion Models
Taesung Kwon*, Jonghyun Park*, Hyungjin Chung†, Jong Chul Ye†
NeurIPS 2026

[C28] Stitched Value Model for Diffusion Alignment
Hyojun Go, Hyungjin Chung†, Prune Truong, Goutam Bhat, Li Mi, Zhaochong An, Zixiang Zhao†, Dominik Narnhofer, Serge Belongie, Federico Tombari, Konrad Schindler
NeurIPS 2026

[C27] Exact Posterior Score Estimation for Solving Linear Inverse Problems
Abbas Mammadov, Ozgur Kara, Kaan Oktay, Iskander Azangulov, Adil Kaan Akan, Hyungjin Chung, James Matthew Rehg, Yee Whye Teh
NeurIPS 2026

[C26] Generating Human Motion Videos using a Cascaded Text-to-Video Framework
Hyelin Nam, Hyojun Go, Byeongjun Park, Byung-Hoon Kim†, Hyungjin Chung†

BMVC 2026

[C25] Inversecrafter: Efficient video recapture as a latent domain inverse problem

Yeobin Hong*, Suhyeon Lee*, Hyungjin Chung†, Jong Chul Ye†

ECCV 2026

[C24] ReDesign: Recovering Editable Design Structures from Raster Images via Agentic Decomposition

Jooyeol Yun, Jintae Park, Hyesu Lim, Junha Hyung, Hyungjin Chung, Jaegul Choo

ECCV 2026

[C23] ContrastiveCFG: Guiding Diffusion Sampling by Contrasting Positive and Negative Concepts

Jinho Chang, Changsun Lee, Hyungjin Chung, Jong Chul Ye

ICML 2026

[C22] ReDirector: Robust Video Diffusion with Reversible Frame Reordering

Byeongjun Park, Byung-Hoon Kim, Hyungjin Chung†, Jong Chul Ye†

CVPR 2026

[C21] Video Parallel Scaling: Aggregating Diverse Frame Subsets for VideoLLMs

Hyungjin Chung, Hyelin Nam, Jiyeon Kim, Hyojun Go, Byeongjun Park, Junho Kim, Joonseok Lee, Seongsu Ha, Byung-Hoon Kim

CVPR 2026 Findings

[C20] A Foundational Brain Dynamics Model via Stochastic Optimal Control

Joonhyeong Park*, Byoungwoo Park*, Chang-Bae Bang, Jungwon Choi, Hyungjin Chung, Byung-Hoon Kim†, Juho Lee†

ICLR 2026

[C19] InvFusion: Bridging Supervised and Zero-shot Diffusion for Inverse Problems

Noam Elata*, Hyungjin Chung*, Jong Chul Ye, Tomer Michaeli, Michael Elad

NeurIPS 2025

[C18] Lesion-Aware Post-Training of Latent Diffusion Models for Synthesizing Diffusion MRI from CT Perfusion

Junhyeok Lee, Hyungwoong Kim, Hyungjin Chung, Heeseong Eom, Jang Joon, Chul-Ho Sohn, Kyu Sung Choi

MICCAI 2025

[C17] CapeLLM: Support-Free Category-Agnostic Pose Estimation with Multimodal Large Language Models

Junho Kim, Hyungjin Chung†, Byung-Hoon Kim†

ICCV 2025

[C16] VideoRFSplat: Direct Scene-Level Text-to-3D Gaussian Splatting Generation with Flexible Pose and Multi-View Joint Modeling

Hyojun Go*, Byeongjun Park*, Hyelin Nam, Byung-Hoon Kim, Hyungjin Chung†, Changick Kim†

ICCV 2025

[C15] SteerX: Creating Any Camera-Free 3D and 4D Scenes with Geometric Steering

Byeongjun Park*, Hyojun Go*, Hyelin Nam, Byung-Hoon Kim, Hyungjin Chung†, Changick Kim†

ICCV 2025

[C14] Derivative-Free Diffusion Manifold-Constrained Gradient for Unified XAI

Won Jun Kim*, Hyungjin Chung*, Jemin Kim*, Byeongsu Sim, Sangmin Lee, Jong Chul Ye

CVPR 2025

[C13] CFG++: Manifold-constrained Classifier Free Guidance for Diffusion Models

Hyungjin Chung*, Jeongsol Kim*, Geon-Yeong Park*, Hyelin Nam*, Jong Chul Ye

ICLR 2025

[C12] Regularization by texts for latent diffusion inverse solvers
Jeongsol Kim*, Geon-Yeong Park*, Hyungjin Chung, Jong Chul Ye
ICLR 2025 (spotlight)

[C11] Deep Diffusion Image Prior for Efficient OOD Adaptation in 3D Inverse Problems
Hyungjin Chung and Jong Chul Ye
ECCV 2024

[C10] Prompt-tuning Latent Diffusion Models for Inverse Problems
Hyungjin Chung, Jong Chul Ye, Peyman Milanfar, Mauricio Delbracio
ICML 2024

[C9] Decomposed Diffusion Sampler for Accelerating Large-Scale Inverse Problems
Hyungjin Chung, Suhyeon Lee, Jong Chul Ye
ICLR 2024

[C8] Direct Diffusion Bridge using Data Consistency for Inverse Problems
Hyungjin Chung, Jeongsol Kim, Jong Chul Ye
NeurIPS 2023

[C7] Improving 3D Imaging with Pre-Trained Perpendicular 2D Diffusion Models
Suhyeon Lee*, Hyungjin Chung*, Minyoung Park, Jonghyuk Park, Wi-Sun Ryu, Jong Chul Ye
ICCV 2023

[C6] Score-based Diffusion Models for Bayesian Image Reconstruction
Michael T. Mccann, Hyungjin Chung, Jong Chul Ye, Marc L. Klasky
ICIP 2023

[C5] Parallel Diffusion Models of Operator and Image for Blind Inverse Problems
Hyungjin Chung*, Jeongsol Kim*, Sehui Kim, Jong Chul Ye
CVPR 2023

[C4] Solving 3d inverse problems using pre-trained 2d diffusion models
Hyungjin Chung*, Dohoon Ryu*, Michael T. Mccann, Marc L. Klasky, Jong Chul Ye
CVPR 2023

[C3] Diffusion Posterior Sampling for General Noisy Inverse Problems
Hyungjin Chung*, Jeongsol Kim*, Michael T. Mccann, Marc L. Klasky, Jong Chul Ye
ICLR 2023 (Notable-top-25%)

[C2] Improving Diffusion Models for Inverse Problems using Manifold Constraints
Hyungjin Chung*, Byeongsu Sim*, Dohoon Ryu, Jong Chul Ye
NeurIPS 2022

[C1] Come-Closer-Diffuse-Faster: Accelerating Conditional Diffusion Models for Inverse Problems through Stochastic Contraction
Hyungjin Chung, Byeongsu Sim, and Jong Chul Ye
CVPR 2022

Journal publications

[J16] ContextMRI: Enhancing Compressed Sensing MRI through Metadata Conditioning
Hyungjin Chung*, Dohun Lee*, Zihui Wu, Byung-Hoon Kim, Katie Bouman, Jong Chul Ye
Medical Image Analysis, 2026

[J15] Deep Learning for Deep Learning Performance: How Much Data Is Needed in Biomedical Imaging?
Kyu Sung Choi, Junhyeok Lee, Hyungjin Chung, Jeong-Hoon Lee
PLOS One, 2026

[J14] Label-independent Framework for Objective Evaluation of Cosmetic Outcome in Breast Cancer

Sangjoon Park, Yong Bae Kim, Jee Suk Chang, Seo Hee Choi, Hyungjin Chung, Ik Jae Lee, Hwa Kyung Byun

Artificial Intelligence in Medicine, 2025

[J13] Steerable Conditional Diffusion for Out-of-Distribution Adaptation in Medical Image Reconstruction

Alexander Denker*, Riccardo Barbano*, Hyungjin Chung*, Tae Hoon Roh, Simon Arrdige, Peter Maass, Bangti Jin, Jong Chul Ye

IEEE TMI, 2025

[P2] Objective and Interpretable Breast Cosmesis Evaluation with Attention Guided Denoising Diffusion Anomaly Detection Model

Sangjoon Park, Yong Bae Kim, Jee Suk Chang, Seo Hee Choi, Hyungjin Chung, Ik Jae Lee, Hwa Kyung Byun

IJROBP, 2024

[J12] Fundus image enhancement through direct diffusion bridges

Sehui Kim*, Hyungjin Chung*, Se Hie Park, Eui-Sang Chung, Kayoung Yi, Jong Chul Ye

IEEE JBHI, 2024

[J11] MR Image Denoising and Super-Resolution Using Regularized Reverse Diffusion

Hyungjin Chung, Eun Sun Lee, Jong Chul Ye

IEEE TMI, 2022

[J10] Low-dose sparse-view HAADF-STEM-EDX tomography of nanocrystals using unsupervised deep learning

Eunju Cha*, Hyungjin Chung*, Jaeduck Jang, Junho Lee, Eunha Lee, Jong Chul Ye

ACS Nano, 2022

[J9] Score-based diffusion models for accelerated MRI

Hyungjin Chung and Jong Chul Ye

Medical Image Analysis, 2021

[J8] Unsupervised Deep Learning Methods for Biological Image Reconstruction and Enhancement

Mehmet Akçakaya, Burhaneddin Yaman, Hyungjin Chung, Jong Chul Ye

IEEE SPM, 2021

[J7] A Deep Learning Model for Diagnosing Gastric Mucosal Lesions Using Endoscopic Images: Development, Validation, and Method Comparison

Joon Yeul Nam*, Hyungjin Chung*, Kyu Sung Choi*, Hyuk Lee* et al.

Gastrointestinal Endoscopy, 2021

[J6] Feature Disentanglement in generating three-dimensional structure from two-dimensional slice with sliceGAN

Hyungjin Chung, Jong Chul Ye

Nature Machine Intelligence, 2021

[J5] Missing Cone Artifacts Removal in ODT using Unsupervised Deep Learning in Projection Domain

Hyungjin Chung*, Jaeyoung Huh*, Geon Kim, Yong Keun Park, Jong Chul Ye

IEEE TCI, 2021

[J4] Two-Stage Deep Learning for Accelerated 3D Time-of-Flight MRA without Matched Training Data

Hyungjin Chung, Eunju Cha, Leonard Sunwoo, Jong Chul Ye

Medical Image Analysis, 2021

[J3] Deep learning STEM-EDX tomography of nanocrystals

Yoseob Han*, Jaeduck Jang*, Eunju Cha*, Junho Lee*, Hyungjin Chung* et al.

Nature Machine Intelligence, 2021 (March Issue cover)

[J2] Unpaired training of deep learning tMRA for flexible spatio-temporal resolution

Eunju Cha, Hyungjin Chung, Eung Yeop Kim, Jong Chul Ye
IEEE TMI, 2021

[J1] Unpaired deep learning for accelerated MRI using optimal transport driven cycleGAN
Gyutaek Oh, Byeongsu Sim, Hyungjin Chung, Leonard Sunwoo, Jong Chul Ye
IEEE TCI, 2020

Books

[B2] Generative Machine Learning Models in Medical Image Computing
Chapter 7: Diffusion Models for Inverse Problems in Medical Imaging
Hyungjin Chung, Jong Chul Ye

[B1] Deep Learning for Biomedical Image Reconstruction
Chapter 12: Image Synthesis in Multi-Contrast MRI with Generative Adversarial Networks
Tolga Çukur, Mahmut Yurt, Salman Ul Hassan Dar, Hyungjin Chung, Jong Chul Ye

Awards

31st Samsung Humantech Silver Award	2025.02
Google Conference Scholarship	2024.05
30th Samsung Humantech Gold Award	2024.02
29th Samsung Humantech Gold Award	2023.02
2020-2024 BISPL Best Researcher Award	2020-2024.12

Professional Service

Area Chair

ICLR	2027-

Committee Member

KSIAM Imaging Science	2026-2027
Technical Group on Image Processing, IEIE	2026-

Grant Reviewer

The Royal Society (Leverhulme Trust Senior Fellowship)	2026

Reviewer (Conference)

ICML (*Gold reviewer 2026*)	2024-
ICLR	2024-
NeurIPS	2022-
NeurIPS Datasets&Benchmarks	2023-
CVPR	2023-
ECCV	2022-
ICCV	2023-
AAAI	2025-
SIGGRAPH	2026-
SIGGRAPH Asia	2025-
MICCAI	2022-

Reviewer (Journal)

NEJM AI
Nature Communications
npj Digital Medicine
Medical Image Analysis
TMLR
IEEE TMI (*Gold Distinguished reviewer 2024*, *Bronze Distinguished reviewer 2023*)
IEEE TPAMI, TCI, TSP, TIP, SPS, SPL, TRPMS
See full list

Invited talks

Future Applications of Generative AI

- *CSE Colloquium, Korea University* 2026.10
- *SKKAI, Sungkyunkwan University* 2026.10

Towards Controllable Video World Models

- *Conference on Artificial Intelligence and Signal Processing* 2026.09

Closing the loop of alignment in Generative AI

- *KAIST EE* 2026.05
- *Spring Workshop on Broadcasting and Media Technology, KIBME* 2026.05

Towards exact posterior sampling with diffusion models

- *KSIAM* 2026.05
- *National Institute for Mathematical Sciences (NIMS)* 2026.05

Test-time Adaptation in Diffusion Models for Medical Image Reconstruction

- *Symposium on AI and Reconstruction for Biomedical Imaging (SyneRBI)* 2026.03

Physically Plausible Generative AI

- *KAIST AI* 2026.03
- *DGIST AI* 2026.03
- *ETH-PRS* 2026.02
- *Joint Winter School on Image Understanding and Image Processing by IEIE* 2026.01

Towards motion understanding and generation in videos

- *Dept. of Applied Statistics and Data Science, Yonsei University* 2025.10
- *Korean Society of Digital Health* 2025.09

Texts in inverse problem solving using diffusion models

- *University of Michigan* 2024.10

Tutorial on Denoising Diffusion Model: Fundamentals & Applications

- *IEIE: Winter School on Biomedical Signal Processing* 2024.02

Adapting diffusion models for inverse problems

- *UCLA, Caltech: Grundfest Memorial Lecture Series in Graphics and Imaging* 2024.02
- *2023 NeurIPS Workshop on diffusion models* 2023.12
- *Google Research* 2023.10

Advances in diffusion models and their applications to inverse problems

- *Guest Lecture, Korea University* 2023.11

Generative (diffusion) models for medical imaging

- *Korean Society for Bone and Mineral Research* 2026.06
- *KoSAIM 2025 summer school* 2025.08
- *International Congress on Magnetic Resonance Imaging (ICMRI) 2023* 2023.11
- *Michigan State University* 2023.09
- *Stanford MedAI* 2023.08
- *MGH, School of Medicine, Harvard University* 2023.08
- *BRIC academic webinar* 2023.03
- *45th meeting, The Korean Society of Abdominal Radiology, 2022* 2022.06

Diffusion models: foundations and applications in biomedical imaging

- *IEEE International Symposium on Biomedical Imaging (ISBI) 2023* 2023.05

Diffusion models for inverse problems

- *POSTECH CSE* 2026.10
- *Joint Workshop on ML for Urban Environmental Modeling and Flows* 2026.09
- *The Korean Institute of Broadcast and Media Engineers* 2025.12
- *University of Minnesota* 2025.11
- *LANL* 2024.11
- *IPA seminar, Korea University* 2024.09
- *Krafton AI* 2024.09
- *DRGem* 2024.08

- *LG AI Research* 2024.08
- *Twelve Labs* 2024.06
- *AI SEOUL 2024* 2024.02
- *Inference & control group seminar, Donders Institute, Radboud Univ.* 2023.01
- *LANL T-CNLS seminar, 2022* 2022.08

Preprints

[P7] Advancing Ultra Low-Field MRI with Synthetic Data and Deep Learning-Based Image Enhancement for Brain Volume Analysis
Peter Hsu, Elisa Marchetto, Hyungjin Chung, Dohun Lee, Jong Chul Ye, Daniel Sodickson, Jelle Veraart, Patricia Johnson

[P4] ACDC: Autoregressive coherent multimodal generation using diffusion correction
Hyungjin Chung*, Dohun Lee*, Jong Chul Ye

[P3] A survey on diffusion models for inverse problems
Giannis Daras, Hyungjin Chung, Chieh-Hsin Lai, Yuki Mitsufuji, Jong Chul Ye, Peyman Milanfar, Alexandros G Dimakis, Mauricio Delbracio

[P2] Amortized Posterior Sampling with Diffusion Prior Distillation
Abbas Mammadov*, Hyungjin Chung*, Jong Chul Ye

[P1] Generative AI for Medical Imaging: extending the MONAI Framework
Pinaya *et al.* (Hyungjin Chung: Contributing author)

Patent

US patent application
- Score-based Diffusion Model for Accelerated MRI and Apparatus thereof 2023

Korea patent publication
- Crowd Deep Learning Method of Medical Artificial Intelligence and Apparatus thereof 2025
- Score-based Diffusion Model for Accelerated MRI and Apparatus thereof 2024
- Task-agnostic image processing method and apparatus using transformer and federated split learning 2024
- Tomography image processing method using neural network based on unsupervised learning to remove missing cone artifacts and apparatus therefor 2023
- Two-Stage unsupervised learning method for 3D Time-of-flight MRA reconstruction and the apparatus thereof 2023

Korea patent application
- Accelerating method of conditional diffusion models for inverse problems using stochastic contraction and the apparatus thereof 2021
- Extreme condition reconstruction method HAADF-STEM-EDX tomography using unsupervised deep learning and the apparatus thereof 2021

References

Byung-Hoon Kim 2024.08 — 2025.08
CIO (EverEx) egyptdj@yonsei.ac.kr
Jong Chul Ye 2019.03 — 2025.02
Ph.D. advisor (KAIST) jong.ye@kaist.ac.kr
Mauricio Delbracio 2023.07 — 2023.11
Host (Google) mdelbra@google.com
Peyman Milanfar 2023.07 — 2023.11
Host (Google) milanfar@google.com
Michael T. McCann 2022.06 — 2022.08
Host (LANL) mccann@lanl.gov